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MILITARY SPECIFICATION

SWITCH, HYDROSTATIC, MARK 24, MODS 0, 1, 2, AND 3

(FOR EXPLODER MECHANISM MARK 19 SERIES)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. This specification states the requirements for the procurement of Hydrostatic Switch MK 24 Mods 0, 1, 2, and 3, which is a single-pole, single-throw, pressure-responsive electric switch with normally closed contacts. It is used to discharge a 4.0 microfarad (mfd) capacitor having an initial charge of 150 volts direct current (dc) down to 0.2 volt dc through a 10 ohm resistor.

1.2 Classification. The hydrostatic switch described in this specification shall be of one class and one grade but of the following types:

Type I - Hydrostatic Switch MK 24 Mod 0
Type II - Hydrostatic Switch MK 24 Mod 1
Type III - Hydrostatic Switch MK 24 Mods 2 and 3

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 Preservation, Methods of
MIL-E-17555 Electronic and Electrical Equipment and Associated Repair Parts; Preparation for Delivery of

FSC 1355

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MIL-T-18404	Torpedoes, Environmental Requirements, General Specifications for

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

NAVAL ORDNANCE SYSTEMS COMMAND

LD 284845	Hydrostatic Switch MK 24 Mods 0, 1, and 2
LD 620098	Hydrostatic Switch MK 24 Mod 3
DWG 1577382	Switch Base Assembly

PUBLICATIONS

NAVAL ORDNANCE SYSTEMS COMMAND

WR-43	Preparation of Quality Assurance Provisions (Including Classification of Characteristics)
OS 6341	Miscellaneous General Design Requirements

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATIONS

Handbook H28	Screw-Thread Standards for the Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402)

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3. REQUIREMENTS

3.1 Description. Hydrostatic Switch MK 24 shall be a pressure-responsive electric switch having normally closed contacts, and shall be composed of a fluid actuating system, a spring-loaded plunger, and an adjusting screw assembly. The housing of the switch shall make electrical contact with the metallic cavity in which it is installed. The second electrical terminal shall be of the male plug type which engages a receptacle at the bottom of the switch cavity. One type of hydrostatic switch differs from another in the amount of pressure required to actuate the switch.

3.2 General requirements

3.2.1 Preproduction samples. Unless otherwise specified in the contract or order, preproduction samples of the switch shall be manufactured using the methods and procedures proposed for the production. The sample will be tested as specified in Section 4 herein and is for the purpose of determining, prior to production, that the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract.

3.2.2 Materials. Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the hydrostatic switch shall conform strictly to the detailed requirements indicated on the drawings and specifications listed in LD 284845 or LD 620098.

3.2.3 Finishes, coating, construction, and dimensions. Finishes, coating, construction, and dimensions for the hydrostatic switch shall be strictly in accordance with the applicable drawings, specifications, and publications listed in LD 284845 or LD 620098.

3.2.4 Moisture, fungus, and corrosion. The hydrostatic switch shall be rendered moisture, fungus, and corrosion resistant as required by OS 6341 and in accordance with detail requirements specified in the drawings and specifications listed in LD 284845 or LD 620098.

3.2.5 Marking. Marking of assemblies, subassemblies, and parts shall be in accordance with the applicable drawings listed in LD 284845 or LD 620098.

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3.2.6 Interchangeability. Component parts of assembled units (unless otherwise specified on the applicable drawings) fabricated in strict agreement with their applicable drawings shall be interchangeable and shall assemble with mating components of the unit without selection or fitting. If, during assembly of the fabricated parts of a unit, selection or fitting is required and changes in dimensions are necessary, no changes in the parts as delineated on applicable drawings will be made without prior revision and correction of the applicable drawings. See 3.2.9.

3.2.7 Gages. The contractor shall provide himself with whatever gages are necessary and adequate to ensure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings. If so stipulated in the contract, order, or requisition, the Government will furnish drawings of pertinent Navy Final Inspection Gages for guidance in the design of the contractor's inspection gages. However, such procedure shall not serve to relieve the contractor of his responsibility in the design and manufacture of such gages as may be required for the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Navy Final Inspection Gages.

3.2.8 Threads. Unless otherwise specified, all threads shall be in accordance with National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings.

3.2.9 Conflicting requirements. Conflicting requirements arising between this specification or any specifications, publications, or drawings listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation and clarification.

3.3 Definitions. The environmental conditions referred to in this specification are defined in MIL-T-18404

3.4 Performance requirements and product characteristics. The hydrostatic switch shall meet the following performance requirements and product characteristics:

3.4.1 Loosening torque. The threaded retaining plate shall remain securely fixed in place under a torque of 3 inch-pounds applied in the loosening direction.

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3.4.2 Oil leakage. The diaphragm and adjacent parts shall evidence no leakage of oil after the face of the switch is subjected to a water pressure of 750 pounds per square inch gage (psig).

3.4.3 Contact staking. The staking of the contact assembly in the adjusted position shall resist a torque of 10 inch-pounds applied in either direction, and shall not change the pressure adjustment beyond the tolerance specified in 3.4.4.

3.4.4 Sensitivity

3.4.4.1 Types I and II. After the contact assembly has been adjusted to contact open position, in accordance with the applicable hydrostatic pressure settings in Table I, and the pressure is increased to 750 ± 25 psig, a reduction in pressure shall cause the switch to close before dropping below the values specified in Table I. After adjustment, the switch shall not vary from the prescribed setting when subjected to ten consecutive pressure cycles of from 0 to 750 ± 25 psig. The increasing or decreasing hydrostatic pressure shall be applied at an approximate rate of 2 psi per second between 0 and 50 psig, and 25 to 40 psi per second between 50 and 750 psig.

TABLE I

Type No.	Mod No.	Pressure Setting
I	0	22 ± 2 psig (dynamic)
II	1	15 ± 2 psig (dynamic)
III	2 & 3	18 ± 1.5 psig (static) 13 psig min. (dynamic)

3.4.4.2 Type III.

3.4.4.2.1 Static sensitivity. After the contact assembly has been adjusted to contact open position, in accordance with the applicable hydrostatic pressure settings in Table I, and the pressure is increased to 750 ± 25 psig, a reduction in pressure shall cause the switch to close before dropping below the value specified in Table I. After adjustment, the switch shall not vary from the prescribed setting when subjected to ten consecutive pressure cycles of from 0 to 750 ± 25 psig. The increasing hydrostatic pressure shall be applied at a rate of 0.2 psi per second from 10 to 30 psig and thereafter at a rate of 25 to 40 psi per second up to a reading of 750 ± 25 psig.

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3.4.4.2.2 Dynamic sensitivity. With the switch subjected to a hydrostatic pressure decreasing at a rate of 6 psi per second from 30 to 10 psig, the contacts at the switch shall close to a contact resistance of one ohm maximum at 13 psig minimum.

3.4.5 Dielectric strength. When the switch contacts are open, the insulation resistance between the male terminal and the housing shall be not less than 10 megohms under an applied potential of 500 volts dc.

3.4.6 Contact resistance. When the switch contacts are closed, the electrical resistance between the male terminal and the housing shall be not more than 0.2 ohm at 0 psig.

3.4.7 Operating life. The hydrostatic switch shall have an operating life of not less than 100 complete cycles, as defined in 4.4.7.1.

3.5 Environmental requirements. The hydrostatic switch shall meet the other applicable requirements of this specification after or while being subjected to environmental conditions defined below, when tested as specified in Section 4.

3.5.1 Vibration

3.5.1.1 Vibration; packaged and packed. The hydrostatic switch packed and packaged as defined in Section 5, shall be tested in accordance with 4.4.1.1 and thereafter shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.5.1.2 Vibration; ready nonoperating or ready condition. The hydrostatic switch, less pack or packaging, shall be capable of withstanding vibrations of 3g's over the frequency range 10 to 100 cps in each of three mutually perpendicular planes for time periods and temperatures as specified in 4.4.1.2. Thereafter, the switch shall show no damage and shall meet all performance requirements and product characteristics of this specification.

3.5.1.3 Vibration; operating condition. The hydrostatic switches, shall be capable of withstanding vibrations of 1g over the frequency range of 20 to 150 cps in each of three mutually perpendicular directions for time periods and temperatures as specified in 4.4.1.3. While undergoing and after the vibration test, the switch shall show no damage and shall meet all performance requirements and product characteristics of this specification.

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3.5.2 Shock

3.5.2.1 Shock; packaged and packed. The hydrostatic switch, packaged and packed as specified in Section 5, shall be tested in accordance with 4.4.2.1 and thereafter shall show no damage and shall meet the performance requirements and product characteristics of this specification.

3.5.2.2 Shock; operating condition. The hydrostatic switch shall be capable of withstanding shock pulses of 150g's peak magnitude, not less than 8 ms pulse duration, and approximating a one-half sine wave pulse, as specified in 4.4.2.2. Thereafter, the switch shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.5.3 Acceleration. The hydrostatic switch shall be capable of withstanding acceleration of 15 g's applied for not less than 0.2 second, as specified in 4.4.3. Thereafter, the switch shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.5.4 Temperature and humidity.

3.5.4.1 Transportation and storage condition. The hydrostatic switch, packaged and packed as defined in Section 5, shall be subjected to the temperature and humidity conditions specified in 4.4.4.1, and thereafter shall show no corrosion or damage and shall meet all the performance requirements and product characteristics of this specification.

3.5.4.2 High and low temperature operation. The hydrostatic switch, less pack and packaging and mounted as specified in 4.4.4.2 and 4.4.4.3, shall be capable of withstanding temperatures over the range of -40°F to +130°F as specified in 4.4.4.2 and 4.4.4.3. At the conclusion of the test, and while still at the test temperature, the switch shall show no corrosion or damage and shall be capable of meeting all the performance requirements and product characteristics of this specification.

3.5.5 External air pressure. The hydrostatic switch shall be subjected to pressure of 500 psia as specified in 4.4.5. Thereafter, the switch shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

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3.5.6 Salt spray test. The hydrostatic switch shall be subjected to salt-laden atmosphere as specified in MIL-STD-202, Method 101B. Thereafter, the switch shall show no corrosion or damage and shall meet all the performance requirements and product characteristics of this specification.

3.5.7 Storage life. The hydrostatic switch, packed and packaged in accordance with 5.1.1, shall be operable and shall meet all the performance requirements and product characteristics of this specification after being stored for a period of at least eight years under temperature and humidity conditions as follows:

3.5.7.1 Temperature: Minus 65 degrees F to plus 160 degrees F (minus 53.89 degrees C to plus 71.11 degrees C).

3.5.7.2 Humidity: Up to 95 percent relative (maximum).

3.5.7.3 To meet the storage life requirement, rubber O-rings and rubber diaphragms shall be replaced after every five years of continuous storage.

3.6 Cleanup. Prior to and after final assembly, all components shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. Burrs and sharp edges as well as rosin flash shall be removed.

3.7 Workmanship. The hydrostatic switch, including all parts and accessories, shall be constructed and finished in a manner to ensure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, marking of parts and assemblies, plating, riveting, machine screw assemblage, staking, and freedom of parts from burrs and sharp edges.

The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified,

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the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure that supplies and services conform to prescribed requirements.

4.1.1 Classification of inspections. Inspection of the hydrostatic switch shall be classified as follows:

a. Preproduction inspection (paragraph 4.4)
b. Acceptance inspection (paragraph 4.5)
c. Periodic production inspection (paragraph 4.4)

4.2 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as Critical, Major, or Minor, in accordance with WR-43. Tests and examinations that verify critical characteristics are identified by the Symbol (C), and those that verify major characteristics are identified by the symbol (M). The number following the classification symbol indicates the serial number of the test or examination. Tests and examinations that are not annotated with a classification code are classified minor.

4.3 Sampling

4.3.1 Preproduction sample. A preproduction sample of five switches manufactured in accordance with 3.2.1 shall be subjected to the preproduction tests detailed in 4.4 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the switches by the supplier, prior to the approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Periodic production sampling. Periodic production samples shall be selected from units that have satisfactorily passed the acceptance inspections of 4.5. These samples shall be subjected to the tests of 4.4 at an activity designated by the procuring activity. Periodic production sampling shall be as follows:

Step 1. Upon initiation of production randomly select one unit from each group of 10 units. Upon satisfactory compliance of samples selected from 3 consecutive groups, shift to Step 2.

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Step 2. Randomly select one unit from each group of 25 units. Upon satisfactory compliance of samples selected from 3 consecutive groups, shift to Step 3.

Step 3. Randomly select one unit from each group of 50 units. Continue Step 3 for the balance of the contract or until a defective unit is found.

When a defective unit is found in any group, acceptance of all units on hand shall be withheld and the contractor shall:

a. Determine the cause of the defect.

b. Take the necessary actions to correct or eliminate the defect from all units on hand and, if so directed by the procuring activity, from those units previously shipped.

c. Submit to the procuring activity for approval the proposed corrective action to prevent recurrence of the defect.

Return to Step 1 and proceed as indicated when inspection is resumed.

4.3.3 Acceptance sampling. The tests and examinations detailed in 4.5 shall be conducted on each hydrostatic switch to be delivered; therefore, there shall be no sampling for acceptance. Defective units shall be rejected.

4.4 Preproduction and periodic production inspection (M101). The preproduction and periodic production samples, after satisfactorily passing the acceptance inspections detailed in 4.5, shall be subjected to the following tests and examinations. Certified inspection data covering the results of preproduction and periodic production inspections shall be forwarded to the Naval Ordnance Systems Command. Failure of any switch to satisfy all requirements of this specification shall be considered cause for rejection.

4.4.1 Vibration tests

4.4.1.1 Vibration; packaged and packed. The switch, packaged and packed in accordance with Section 5 or the contract or purchase order, shall be vibrated on a barrier equipped platform of a vibrator. Staring at 2 cps, gradually increase frequency until pack just begins to leave the table. At this point, vibrate the pack for a period of one hour. Thereafter the switch shall show no evidence of damage and shall meet all the performance requirements of Section 3.

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4.4.1.2 Vibration; ready nonoperating or ready condition. The hydrostatic switch, mounted as shown on Dwg 1577382, shall be subjected to the vibration testing of Schedule I in each of three mutually perpendicular axes. Thereafter, the switch shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

Schedule I

Frequency Range	Amplitude	Minimum Test Duration		
		Room Temperature	-20°F	+130°F
10-30 cps	0.060 ± 0.006 in.	60 min.	15 min.	15 min.
31-60 cps	3 g ± 0.3 g	60 min.	15 min.	15 min.
61-100 cps	3 g ± 0.3 g	60 min.	15 min.	15 min.

The frequency shall be continuously varied between each of the limits given, at a uniform rate over a three to five minute period, for the duration of the test.

As an alternate method, the hydrostatic switch shall be vibrated at specific frequencies differing by not more than five cps in each of the test axes and covering the entire frequency range and amplitude of Schedule I. The total test time shall be not less than that of Schedule I, and the times at the specific frequencies shall be equally divided.

4.4.1.3 Vibration; operating condition. The hydrostatic switch, mounted as shown on Dwg 1577382, shall be subjected to the vibration tests of Schedule II in each of three mutually perpendicular axes. During and thereafter, the switch shall meet all the performance requirements and product characteristics of this specification.

Schedule II

Frequency Range	Amplitude	Minimum Test Duration
		Room Temperature
20-60 cps	1 g ± 0.1 g	15 min.
61-150 cps	1 g ± 0.1 g	15 min.

The frequency shall be continuously varied between each of the limits given, at a uniform rate over a three to five minute period, for the duration of the test.

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As an alternate method, the hydrostatic switch shall be vibrated at specific frequencies differing by not more than five cps in each of the test axes and covering the entire frequency range and amplitudes of Schedule II. The total test time shall be not less than that of Schedule II, and the test times at the specific frequencies shall be equally divided.

4.4.2 Shock tests

4.4.2.1 Shock; packaged and packed. The switch, packaged and packed in accordance with Section 5 of the contract or purchase order, shall be dropped from a free fall height of 36 inches in accordance with the requirements of WR-11 onto an essentially nonyielding surface. Thereafter the switch shall show no evidence of damage and shall be capable of meeting the performance requirements of Section 3.

4.4.2.2 Shock; operating condition. The hydrostatic switch, mounted as shown on Dwg 1577382, shall be subjected to three shock pulses of not less than 8 ms duration and approximating a one-half sine wave. The pulse shall have a magnitude of 150 ± 15 g's and shall be applied in both directions along each of three mutually perpendicular axes (total of 18 shock pulses). Thereafter, the switch shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

4.4.3 Acceleration. The hydrostatic switch, mounted as shown on Drawing 1577382, shall be subjected to 15 ± 2 g's acceleration for a period of not less than 0.2 second. This test shall be conducted in three mutually perpendicular axes, with the inertial load applied in both directions.

4.4.4 Temperature and humidity tests.

4.4.4.1 Transportation and storage temperature test. The hydrostatic switch, when inside of its package, shall be subjected to temperature and humidity cycling as specified for a nonoperating torpedo in MIL-T-18404.

4.4.4.2 High temperature. The hydrostatic switch shall be placed in a test chamber whose temperature shall be raised to + 130 ± 5° F for a period of 48 ± 5 hours. At the end of this test period, and while still at this temperature, the switch shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

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4.4.4.3 Low temperature. The hydrostatic switch shall be placed in a test chamber whose temperature shall be lowered to - 40 ± 5°F for a period of 48 ± 5 hours. At the end of this test period, and while still at this temperature, the switch shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

4.4.5 External air pressure. The hydrostatic switch shall be subjected to an external pneumatic pressure of 500 ± 25 psia for a 30-minute minimum period. Thereafter, the switch shall show no damage and shall meet all the performance requirements and product characteristics of this specification. Only that portion of the switch above the O-ring shall be exposed to the test condition.

4.4.6 Salt spray test. Conduct salt spray test in accordance with MIL-STD-202, Method 101B, Test Condition B. Thereafter, the hydrostatic switch shall meet all the performance requirements and product characteristics of this specification. Only that portion of the switch above the O-ring shall be exposed to the test condition.

4.4.7 Operating life.

4.4.7.1 Definition of one cycle. One cycle of operation for Hydrostatic switch MK 24 shall be defined as the application of water pressure to the face of the switch throughout the interval from zero psig to 750 psig and back to zero again at the rates specified in 4.5.4.4.1. When the switch contacts open, a 4.0 mfd plus or minus 10 percent capacitor shall be charged from a dc power source with 9000 ohms plus or minus 5 percent output resistance to 145 plus or minus 5 volts dc. When the switch contacts close, the same capacitor shall be discharged to 1 volt dc maximum through a 10 ohm plus or minus 5 percent resistor.

4.4.7.2 Operating life test. Mount the switch in the cavity of the test fixture (4.5.3.2) and connect the switch to the electrical circuit (4.4.7.1). Subject the hydrostatic switch to not less than 100 complete cycles under the environmental conditions described in 4.5.1. Switch operation to charge and discharge the capacitor as described shall occur within the operating pressure tolerances of Table I throughout the life test.

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4.5 Acceptance inspection. Each hydrostatic switch delivered under contract shall be subjected to the following tests and examinations. Failure to meet any requirement specified herein shall be considered cause for rejection.

4.5.1 Test conditions. Unless otherwise specified, the hydrostatic switch shall be subjected to acceptance tests under the following conditions:

4.5.1.1 Temperature: Room ambient, plus 65 degrees F to plus 95 degrees F (plus 18.33 degrees C to plus 35.0 degrees C).

4.5.1.2 Altitude: Normal ground.

4.5.1.3 Vibration: None.

4.5.1.4 Humidity: Room ambient to 95 percent relative (maximum).

4.5.2 Test and inspection equipment and facilities. The manufacturer shall furnish and maintain all necessary test equipment and facilities and shall provide personnel for performing all acceptance tests. The test equipment shall be adequate in quantity and, where definite requirements are not specified, shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

4.5.3 Test equipment. The following items of test equipment are required to perform the acceptance tests set forth in this specification:

4.5.3.1 Suitable torque-indicating instruments capable of measuring a 3 inch-pound and a 10 inch-pound torque.

4.5.3.2 A test fixture in the form of a simulated hydrostatic switch cavity to which water pressure may be applied at the following rates in the given ranges:

(a) At 0.2 psi per second from 10 to 30 psig and from 30 to 10 psig.

(b) At 2 psi per second from 0 to 50 psig and from 50 to 0 psig.

(c) At 6 psi per second from 30 to 10 psig.

(d) At 25 to 40 psi per second from 30 to 750 psig and from 750 to 30 psig.

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A female receptacle for the switch is not required at the base of the cavity, and the contact may be simply exposed to the atmosphere. The hydraulic system of the test fixture shall incorporate two water pressure gauges with suggested full scale readings of 50 psig and 1000 psig.

4.5.3.3 A pressure source capable of supplying water pressures up to at least 750 psig.

4.5.3.4 An electrical instrument, or a suitable electrical circuit such as one containing a light bulb indicator, to determine the instant of switch operation.

4.5.3.5 A megohmmeter with an integral source of 500 volts dc and a midscale reading in the order of 10 megohms.

4.5.3.6 An ohmmeter capable of measuring resistances in the order of 0.2 ohm.

4.5.4 Test procedure. The acceptance tests shall be performed in accordance with the following procedures:

4.5.4.1 Loosening torque test (M102). Make sure that the cement has dried on the threaded retaining plate and that all excess cementing residue has been removed, except that remaining in the groove. Using a suitable torque indicating instrument (4.5.3.1), apply a torque of 3 inch-pounds to the plate in the loosening direction. The switch shall meet the requirements of 3.4.1.

4.5.4.2 Oil leakage test (M103). Remove all oil from the exterior surface of the diaphragm, plate, and housing. Mount the switch in the test fixture (4.5.3.2) and apply water pressure to the face of the switch until 750 psig is reached. Maintain this pressure for a period of five minutes. Relieve the water pressure, remove the switch from the fixture, and before drying, examine the diaphragm and adjacent parts for evidence of oil. The switch shall meet the requirements of 3.4.2.

4.5.4.3 Contact staking test (M104). Apply a torque of 10 inch-pounds to the contact adjusting screw after it has been staked in the adjusted position through means of a suitable torque indicating instrument (4.5.3.1). The contact adjusting screw shall resist this torque when applied in either direction, and the operating pressure of the switch, as tested in 4.5.4.4, shall remain within the specified tolerance.

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4.5.4.4 Sensitivity test

4.5.4.4.1 Types I and II (M105). Mount the switch in the cavity of the test fixture (4.5.3.2) and connect the electrical indicating circuit (4.5.3.4) in series with the switch. Apply water pressure (4.5.3.3) to the face of the switch at a rate of 2 psi per second up to a reading of 50 psig and thereafter at a rate of 25 to 40 psi per second up to a reading of 750 psig. With increasing pressure the switch shall operate to open its contacts at the pressure setting specified in 3.4.4.1. Reduce the pressure from 750 psig to 50 psig at a rate of 25 to 40 psi per second and from 50 psig to 0 at a rate of 2 psi per second. With decreasing pressure the switch shall operate to close its contacts at the pressure setting specified in 3.4.4.1. This test shall be performed ten consecutive times on each hydrostatic switch and the operating pressure shall remain within the designated tolerance. Before dismounting the switch from the test fixture, the dielectric test (4.5.4.5) may be performed..

4.5.4.4.2 Type III

4.5.4.4.2.1 Static sensitivity test (M106). Mount the switch in the cavity of the test fixture (4.5.3.2) and connect the electrical indicating switch (4.5.3.4) in series with the switch. Apply water pressure (4.5.3.3) to the face of the switch at a rate of 0.2 psi per second from 10 to 30 psig and thereafter at a rate of 25 psi per second up to a reading of 750 psig. With increasing pressure, the switch shall operate to open its contacts at the pressure setting specified in 3.4.4.2.1. Reduce the pressure from 750 to 30 psig at a rate of 25 psi per second and proceed thereafter to the dynamic sensitivity tests of 4.5.4.4.2.2. These tests, the static sensitivity tests and the dynamic sensitivity tests, shall be performed in conjunction with each other ten consecutive times on each Type III hydrostatic switch. The operating pressures shall remain within the specified tolerance of Table I.

4.5.4.4.2.2 Dynamic sensitivity test (M107). Decrease the pressure at a rate of 6 psi per second from 30 to 10 psig. The contacts of the switch shall close at the pressure specified in 3.4.4.2.2. Before dismounting the switch from the test fixture, the test of 4.5.4.5 may be performed.

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4.5.4.5 Dielectric test (M108). With the switch mounted in the test fixture, apply enough water pressure to open the switch contacts. Disconnect the electrical indicating circuit and connect a megohmmeter (4.5.3.5) between the male terminal and the housing of the switch. The dielectric strength shall meet the requirements of 3.4.5.

4.5.4.6 Contact resistance test (M109). When the switch contacts are in their normally closed position, connect an ohmmeter (4.5.3.6) between the male terminal and the housing of the switch. Contact resistance shall be as specified in 3.4.6.

4.5.4.7 Packaging and packing. Hydrostatic switch packaging and packing shall be examined to ensure that it complies fully with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A. In addition to the following, Level A packaging shall be in accordance with MIL-E-17555.

5.1.1.1 Cleaning, drying, and preservative application. Not applicable.

5.1.1.2 Unit packaging. Unit packaging or the hydrostatic switch, shall be in accordance with Method 1C-2 of MIL-P-116. The unit package shall contain one switch.

5.1.2 Level B. In addition to the following, Level B packaging shall be in accordance with MIL-E-17555.

5.1.2.1 Cleaning, drying, and preservation application. Not applicable.

5.1.2.2 Unit packaging. Same as 5.1.1.2

5.1.3 Level C.

5.1.3.1 Cleaning, drying and preservative application. Not applicable.

5.1.3.2 Unit packaging. Unit packaging shall be in accordance with Method III of MIL-P-116. The unit package shall contain one switch.

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5.2 Packing.

5.2.1 Level A. In addition to the following, Level A packing shall be in accordance with MIL-E-17555.

5.2.1.1 Exterior containers. Unless otherwise specified in the order or contract, the switch, packaged as specified in 5.1.1.2, may be packed in an overseas type container in accordance with MIL-E-17555 (gross weight not to exceed 200 pounds.)

5.2.2 Level B. In addition to the following, Level B packing shall be in accordance with MIL-P-17555.

5.2.1.1 Exterior containers. Unless otherwise specified in the order or contract, the switch, packaged as specified in 5.1.1.2, may be packed in a domestic type container in accordance with MIL-E-17555 (gross weight not to exceed 200 pounds).

5.2.3 Level C. The hydrostatic switch, packaged in accordance with 5.1.1.2, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers shall conform to the carrier rules and regulations applicable to the mode of transportation.

5.3 Level A, B, or C cushioning. The unit package and pack shall incorporate sufficient cushioning material, bracing, or other adequate shock absorbing devices to ensure that the switch will meet all requirements of Section 3 after being tested in accordance with 4.4.1.1 and 4.4.2.1. At the time of packaging or packing, the moisture content of the cushioning material shall be no more than 5%. The pH range of the material shall be between 6.0 and 8.0.

5.4 Marking.

5.4.1 Special markings. None, unless otherwise specified.

5.4.2 Normal markings. In addition to the markings required by contract or order, shipping containers shall be marked in accordance with the requirements of MIL-STD-129. including serial numbers.

6. NOTES

6.1 Intended use. Hydrostatic Switch MK 24 is intended for use in Exploder Mechanism MK 19 series.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Any exceptions to this specification, applicable drawings, or other documents.

(c) Applicable preservation, packaging, and packing if hydrostatic switches are procured as spare or replacement parts.

(d) Type of hydrostatic switch required.

(e) Assigned activity for preproduction and periodic production tests.

(f) That certified inspection data covering the results of preproduction and periodic production tests shall be forwarded to the Naval Ordnance Systems Command when testing is accomplished at other than a Government activity.

(g) All additional samples required because of failure of any unit(s) to pass the prescribed periodic production examinations shall be provided and inspected at the expense of the supplier.

6.3 Criteria for use of proper level of preservation, packaging, and packing shall be as follows:

For Level A. This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

For Level B. This level shall be used only when it is definitely known that the item will be held in covered storage in overseas locations for six months or less or in domestic locations for an indefinite period.

For Level C. This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-S-19820B(OS) SWITCH, HYDROSTATIC, MARK 24 MODS 0, 1, 2, and 3

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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